DOMAIN MODULE

Data Handling

Chapter 5 · Class VIII Mathematics · NCERT

Adaptive Learning System | 5E Constructivist Framework | Three Content Paths

Component	Detail
Pedagogical Model	5E Framework: Engage → Explore → Explain → Elaborate → Evaluate
Pre-Test Design	8 Medium questions — prior knowledge only (not Chapter 5 content)
Path A (0–40%)	FULL foundational content — all concepts, all examples, step-by-step
Path B (40–70%)	STANDARD content — complete concept + focused examples + practice
Path C (70–100%)	ADVANCED content — extended examples, analysis, application tasks
Inline Questions	3–4 per module; differentiated by path; hint + remedial on wrong answer
Post-Test	10 questions — same 13 skill tags; learning gain tracked per sub-topic

Section 1: Pre-Test — 8 Medium Questions

Pre-Test Purpose

These 8 questions test PRIOR KNOWLEDGE only — reading basic graphs, working with fractions and percentages, tally marks, and basic chance. They do NOT test Chapter 5 content.

Results are used to route each learner into Path A, B, or C independently per sub-module.

A learner may be in Path A for histograms but Path C for data organisation — routing is per-module, not per-learner.

Q1 [Medium]

A bar graph shows books read: Anu=4, Raj=7, Mia=5, Tom=3, Sia=6. Who read the MOST books?

(A) Anu
(B) Raj
(C) Mia
(D) Tom

Hint: Tallest bar = highest value.

Answer (B): Raj read 7 — the highest count. Tags: bar_graph_read, Module 2.1

Q2 [Medium]

Pictograph: one symbol = 100 cars. August shows 3 full symbols. How many cars?

(A) 30 cars
(B) 300 cars
(C) 3 cars
(D) 3,000 cars

Hint: Multiply symbols by the value each symbol represents.

Answer (B): 3 × 100 = 300 cars. Tags: pictograph_read, Module 2.1

Q3 [Medium]

Double bar graph: Maths 2005-06=30, 2006-07=70. What is the improvement?

(A) 30 marks
(B) 40 marks
(C) 50 marks
(D) 70 marks

Hint: Improvement = new score − old score.

Answer (B): 70 − 30 = 40 marks. Tags: double_bar_read, Module 2.1

Q4 [Medium]

Out of 20 students, 8 prefer Science. What fraction prefer Science?

(A) 5/20
(B) 8/20
(C) 8/5
(D) 13/20

Hint: Fraction = preferred ÷ total.

Answer (B): 8 out of 20 → 8/20. Tags: fraction_percent, Module 2.3

Q5 [Medium]

Which is the correct tally for 7?

(A) |||| ||

(B) |||| |||

(C) |||||||

(D) |||| |

Hint: Groups of 5 are crossed. 7 = 5+2.

Answer (A): One group of 5 (||||) + 2 singles (||). Tags: frequency_table, Module 2.1

Q6 [Medium]

Spinner: 4 equal sections (Red, Blue, Green, Yellow). Chance of landing on Blue?

(A) 1 out of 2

(B) 1 out of 4

(C) 2 out of 4

(D) 4 out of 4

Hint: Count total sections and blue sections.

Answer (B): 1 blue out of 4 equal → 1 out of 4. Tags: prob_basic, Module 2.4

Q7 [Medium]

Pie chart: 50% prefer Chocolate. If 200 students total, how many prefer Chocolate?

(A) 50

(B) 75

(C) 100

(D) 150

Hint: 50% means half. Half of 200 = ?

Answer (C): 50% of 200 = 100. Tags: fraction_percent, Module 2.3

Q8 [Medium]

Marks: 12, 18, 25, 31, 14. Intervals: 0–10, 10–20, 20–30, 30–40. Where does 25 go?

(A) 10–20

(B) 20–30

(C) 30–40

(D) 0–10

Hint: 25 is between 20 and 30. Boundary belongs to higher class.

Answer (B): 25 falls in 20–30. Tags: class_interval_basics, Module 2.2

Section 2: Adaptive Routing Logic

After the Pre-Test, the system computes a score per sub-module and routes each learner into the appropriate content path. A learner may be in different paths for different modules.

0–40% PATH A	FULL FOUNDATIONAL: Complete content from the very beginning. Every concept defined. Multiple worked examples with step-by-step solutions. All graph types explained from scratch with drawing rules. Real-life analogies provided. Inline questions start at Easy difficulty. Remedial loop for every wrong answer.
40–70% PATH B	STANDARD: Complete concept explanations with focused examples. Students who know basics but need to master the full topic. Worked examples with clear solution steps. Application-level questions. Inline questions start at Medium. One remedial opportunity.
70–100% PATH C	ADVANCED: Builds on existing understanding. Extended and complex examples. Multi-step problems. Connections between topics. Analysis and interpretation tasks beyond simple calculation. Challenge questions unlocked. Inline questions start at Hard.

Sub-Module	Questions Mapped	Both wrong → Path A	1 correct → Path B	Both correct → Path C
2.1 Data Org & Graphs	Q1,Q2,Q3,Q5	All 4 wrong → Path A	2–3 correct → Path B	3–4 correct → Path C
2.2 Histograms	Q5, Q8	Both wrong → Path A	1 correct → Path B	Both correct → Path C
2.3 Pie Charts	Q4, Q7	Both wrong → Path A	1 correct → Path B	Both correct → Path C
2.4 Probability	Q6	Q6 wrong + low total → A	Q6 correct + low total → B	Q6 correct + strong → C

MODULE 2.1: Data Organisation & Types of Graphs

PATH A | Score 0–40% | Full Foundational Content — All Concepts from Scratch

ENGAGE — Why Do We Need to Organise Data?

Opening Scenario — Think About This

Your teacher asks: "Which sport is most popular in Class 8?" Every student calls out at the same time. You hear: Cricket, Football, Cricket, Badminton, Cricket, Football, Tennis, Cricket … Can you tell which sport is most popular? No — it is noise. You need to COLLECT, ORGANISE, and COUNT responses systematically. That is exactly what Data Handling teaches you to do.

EXPLORE — Raw Data

Below is the raw list of favourite subjects from 22 students. Read it once:

Raw Data (as collected — unorganised)

Art, Mathematics, Science, English, Mathematics, Art, English, Mathematics, English, Art, Science, Art, Science, Science, Mathematics, Art, English, Art, Science, Mathematics, Science, Art

Now answer these without re-reading: Which subject is most liked? Which is least liked? How many students like Science? — Almost everyone gives a wrong answer or cannot answer at all. This experience of confusion is precisely WHY we need tally marks and frequency tables.

EXPLAIN — Core Concepts

What Is Data?

Data is a collection of facts, numbers, or observations gathered for a specific purpose. The raw list above is called raw data — it is unorganised. When we process and organise it, it becomes useful information.

What Are Tally Marks?

How Tally Marks Work

Every time a value appears, make ONE stroke: |

After 4 strokes (||||), the 5th stroke crosses them diagonally: |||| — this completes a group of 5.

Tally marks in groups of 5 make counting fast and error-free.

Example: To represent 13 items → |||| |||| ||| (two groups of 5 = 10, plus 3 singles = 13)

Example: To represent 7 → |||| || (one group of 5 + 2 singles)

What Is Frequency?

Frequency is the number of times a particular value (or category) appears in the data. It is simply the COUNT for that category. A Frequency Distribution Table organises data by listing every category with its tally marks and frequency.

Step-by-Step: Building a Frequency Distribution Table

How to Make a Frequency Table

1. List all categories in a column (here: Art, Mathematics, Science, English).
2. Create two more columns: "Tally Marks" and "Frequency".
3. Go through the raw data ONE item at a time, left to right.
4. For each item, add one tally mark in the correct row. Mark it off in the raw data so you do not double-count.
5. Once all data is processed, count the tally marks in each row and write the frequency.

6. Verify: the sum of all frequencies must equal the total number of data items (22 in this case).

Subject	Tally Marks	Frequency	Reading the table
Art	|||| ||	7	7 students chose Art as favourite
Mathematics	||||	5	5 students chose Mathematics
Science	|||| |	6	6 students chose Science
English	||||	4	4 students chose English
TOTAL		22	All 22 students accounted for ✓

Now We Can Answer!

Most popular: Art (frequency 7). Least popular: English (frequency 4). Students who like Science: 6. The table made every question answerable in seconds — that is the power of organising data.

Types of Graphs — Complete Guide

1. Pictograph

A pictograph represents data using pictures or symbols. Each symbol stands for a fixed number of items, shown in a KEY.

Rules for Drawing and Reading a Pictograph

Choose a clear, relevant symbol (e.g., a small car icon for vehicle data).
Decide the SCALE: how many items does one symbol represent? (e.g., 1 symbol = 100 cars)
For values that are not exact multiples, draw a FRACTION of the symbol (half symbol = 50; quarter symbol = 25).
Always include a KEY / LEGEND showing what one full symbol represents.
Label each row with the category name.

Worked Example — Reading a Pictograph

Cars produced per month. Scale: 1 symbol = 100 cars.
January: 2 symbols = 200 cars.
February: 1.5 symbols = 150 cars.
March: 3 symbols = 300 cars.
September: 4 full + 1 half symbols = (4×100) + 50 = 450 cars.
Trick question: if a row shows 3.5 symbols → 3.5 × 100 = 350 cars.

2. Bar Graph

A bar graph represents data using rectangular bars. The HEIGHT of each bar shows the value for that category.

Rules for Drawing a Bar Graph

Horizontal axis (x-axis): list the category names, evenly spaced.
Vertical axis (y-axis): numerical scale, starting from 0, with uniform intervals (e.g., 0, 50, 100, 150 …).
All bars must have EQUAL WIDTH.
GAPS must exist between bars — the bars do not touch. (This shows data is categorical, not continuous.)
The HEIGHT of each bar = the value it represents.
Both axes must be labelled. A title must be given.

Worked Example — Reading a Bar Graph

School enrolment from 2004-05 to 2008-09:

2004-05: 100 students. 2005-06: 200 students. 2006-07: 300 students. 2007-08: 400 students. 2008-09: 500 students.

Q: Which year had maximum enrolment? A: 2008-09 (tallest bar = 500).

Q: How much did enrolment increase from 2003-04 (100) to 2007-08 (400)? A: 400 − 100 = 300 students.

Q: Between which two consecutive years was the increase maximum? Calculate each gap: +100 each year. All equal here.

3. Double Bar Graph

A double bar graph shows TWO datasets side-by-side for the same categories. This allows direct, visual comparison.

How to Read a Double Bar Graph

Two bars appear SIDE-BY-SIDE for each category — one bar for each dataset.
The two bars are coloured or shaded differently to tell them apart.
A LEGEND (colour key) tells you which colour/shade represents which dataset.
To find increase: later bar height − earlier bar height. Positive = increase. Negative = decrease.
To compare: which bar is taller in each pair?

Worked Example — Double Bar Graph

Two years of marks in various subjects. Colours: Blue = 2005-06, Orange = 2006-07.
Maths: 2005-06=30, 2006-07=70. Change = 70−30 = +40 (improved by 40 marks).
Hindi: 2005-06=50, 2006-07=45. Change = 45−50 = −5 (DECREASED by 5 marks).
English: 2005-06=60, 2006-07=70. Change = +10 (improved).
Question: "In which subject did performance decrease?" Answer: Hindi (went from 50 down to 45).

Graph Type	When to Use	Key Feature	Gaps Between Bars?
Pictograph	Small categorical quantities	Symbol = fixed quantity; needs scale key	N/A (uses symbols)
Bar Graph	Categorical data comparison	Equal-width rectangular bars; height = value	YES — always gaps
Double Bar Graph	Compare two datasets across categories	Two coloured bars per category; needs legend	YES — gaps between pairs

PATH B | Score 40–70% | Standard Content — Complete Concepts with Focused Examples and Practice

ENGAGE — The Problem That Graphs Solve

Imagine you have collected 50 responses to "What is your favourite sport?" as a raw list. Reading it takes 2 minutes and you still cannot answer: "Which sport is most popular?" Frequency tables, pictographs, and bar graphs exist to solve this exact problem — to turn a messy list into instant, readable information.

EXPLAIN — Frequency Distribution Tables

What Is a Frequency Table?

A Frequency Distribution Table lists every category in one column, tally marks in the second column, and the frequency (count) in the third. Reading a frequency table is instant — you see the count for any category in one glance.

How Tally Marks Work

Each occurrence of a value is recorded as one stroke |. After four strokes (||||), the fifth stroke crosses them diagonally to form a group of five (||||). Counting in groups of five makes verification fast and avoids errors in long lists.

Subject	Tally Marks	Frequency	Observation
Art	|||| ||	7	Most popular subject
Mathematics	||||	5	Third most popular
Science	||||		6	Second most popular
English	||||	4	Least popular
TOTAL		22	All students accounted for ✓

What the Table Immediately Tells You

Most popular: Art (7). Least popular: English (4). Difference between most and least: 7−4=3. Total responses: 22. None of these required re-reading the raw data.

Pictograph — Reading and Calculating

Key Rules

Rule	Explanation	Example
Scale/Key	One symbol = a fixed quantity. ALWAYS read the key first.	1 car symbol = 100 cars
Full symbol	Value of one complete symbol as per the key	3 symbols = 3 × 100 = 300
Half symbol	Represents half the key value	0.5 symbols = 50 cars
Quarter symbol	Represents one-quarter the key value	0.25 symbols = 25 cars
Reading a row	Count symbols (including fractions) then multiply by key value	4.5 × 100 = 450 cars

Worked Example 1 — Cars Produced

A pictograph shows cars produced. Scale: 1 symbol = 100 cars.

Month	Symbols Shown	Calculation	Cars Produced
January	2 full	2 × 100	200
February	1.5 (1 full + 1 half)	1.5 × 100	150
March	3 full	3 × 100	300
April	2.5 (2 full + 1 half)	2.5 × 100	250
September	4.5 (4 full + 1 half)	4.5 × 100	450

Worked Example 2 — Reverse Calculation

Sometimes you are told the quantity and asked how many symbols to draw.

Example

A shop sold 350 kg of rice. Scale: 1 symbol = 50 kg.
Symbols needed: 350 ÷ 50 = 7 full symbols.
A shop sold 175 kg. 175 ÷ 50 = 3.5 symbols (3 full + 1 half).

Always divide the quantity by the key value to find symbol count.

Bar Graph — Construction and Interpretation

Rules for Drawing a Bar Graph

Rule	What It Means	Common Mistake to Avoid
Equal bar widths	All bars must be the same width	Drawing bars of different widths changes perceived values
Gaps between bars	Leave equal spaces between every bar	No gaps makes it look like a histogram (wrong for categorical data)
y-axis from 0	Scale must start at 0 unless a break mark is shown	Starting y-axis at 50 makes differences look bigger than they are
Uniform scale	Equal numerical intervals on y-axis (e.g., 0,50,100,150)	Unequal intervals (0,10,50,100) distort the graph
Labels on both axes	x-axis: categories. y-axis: frequency or value	Unlabelled axes make a graph unreadable

Worked Example — School Enrolment

Bar graph data: 2003-04=100, 2004-05=200, 2005-06=300, 2006-07=400, 2007-08=450 students.

Question	Working	Answer
Which year had highest enrolment?	Find tallest bar = highest value	2007-08 (450 students)
Increase from 2003-04 to 2006-07?	400 − 100 = 300	300 students
In which year was enrolment double that of 2003-04?	2003-04=100. Double=200. Which year shows 200?	2004-05
Biggest jump between consecutive years?	Check each gap: +100,+100,+100,+50. Max=+100	2003-04 to 2004-05 (or any of first three)

Double Bar Graph — Comparison and Analysis

How to Read a Double Bar Graph

A double bar graph has two bars side-by-side for each category on the x-axis — one for each dataset. A LEGEND identifies which colour/shade belongs to which dataset.

Task	Method
Find increase for a subject	New value − Old value. Positive result = increase.
Find decrease for a subject	New value − Old value. Negative result = decrease.
Find maximum improvement	Calculate change for each subject. Largest positive change = max improvement.
Find which subject decreased	Look for subjects where the second bar is SHORTER than the first.
Compare two subjects in same year	Read both bars for that year and compare heights directly.

Worked Example — Exam Marks Over Two Years

Subject	2005-06	2006-07	Change	Interpretation
Maths	30	70	+40	Biggest improvement
Hindi	50	45	−5	Only subject that DECREASED
Science	65	80	+15	Improved moderately
English	55	60	+5	Slight improvement

Practice Questions from This Graph

1. Which subject showed the maximum improvement? → Maths (+40).2. Which subject decreased? → Hindi (50→45 = −5).3. In which subject did students perform above 70 marks in BOTH years? → Science (65 and 80, both > 60 but only 80 > 70 in 2nd year). Check: only 2006-07 Science (80) and Hindi (50 in 05-06, not >70).4. What is the total improvement across all subjects? → 40+(−5)+15+5 = 55 marks.

PATH C | Score 70–100% | Advanced Content — Multi-Dataset Analysis and Critical Thinking

ELABORATE — Advanced Data Analysis

You have strong existing knowledge of frequency tables, pictographs, bar graphs, and double bar graphs. Path C focuses on multi-dataset comparisons, analytical reasoning, deriving conclusions, and identifying limitations of different graph types.

Concept Extension 1 — When to Use Which Graph

Situation	Best Graph Type	Why	Limitation
Compare one value across many categories	Bar Graph	Height differences are immediately visible	Cannot show two datasets at once
Compare two values across same categories	Double Bar Graph	Side-by-side comparison shows difference clearly	Becomes cluttered with more than 3 datasets
Show quantities using a symbol	Pictograph	Visually engaging; easy to read at a glance	Imprecise for non-multiples; takes space
Count occurrences in raw data	Frequency Table	Exact counts; easy to total and verify	No visual representation
Show a TREND over time	Line Graph (not covered here)	Lines make upward/downward trends obvious	Bar graphs can also show trends but less clearly

Multi-Step Example 1 — Analysing a Double Bar Graph Across Four Classes

A double bar graph shows Science and Maths marks for Classes 6, 7, 8, and 9. Blue = Maths, Orange = Science.

Class	Maths Score	Science Score	Maths − Science	Which Subject Higher?
Class 6	60	55	+5 (Maths)	Maths
Class 7	70	72	−2 (Science)	Science (first overtake)
Class 8	65	75	−10 (Science)	Science
Class 9	80	78	+2 (Maths)	Maths (reversed)

Analytical Questions and Answers

Q1: In which class does Science first overtake Maths? → Class 7 (Science 72 > Maths 70 for the first time).

Q2: Average Maths score across all 4 classes = (60+70+65+80) ÷ 4 = 275 ÷ 4 = 68.75.

Q3: Average Science score = (55+72+75+78) ÷ 4 = 280 ÷ 4 = 70. Science average (70) > Maths average (68.75).

Q4: Maths showed a dip in Class 8 (70→65 then 65→80). What might explain this? → Change in syllabus difficulty, different teacher, etc. (Interpretation, not calculation.)

Q5: If Class 10 follows the trend, estimate Maths: the jumps were +10, −5, +15. Average = +6.7/class. Predict ≈ 80+7 = 87.

Multi-Step Example 2 — Comparing Two Frequency Tables

Class A favourite colours (12 students): Red=4, Blue=5, Green=2, Yellow=1.

Class B favourite colours (12 students): Red=3, Blue=4, Green=3, Yellow=2.

Question	Working	Answer
Most popular across both classes combined?	Blue: 5+4=9. Red: 4+3=7. Green: 2+3=5. Yellow: 1+2=3.	Blue (9 students total)
Which class shows more even distribution?	Class A max=5 min=1 (spread 4). Class B max=4 min=2 (spread 2).	Class B more even
If 2 students leave Class A (both Blue), new Blue%?	New Blue=3 out of 10. 3/10 = 30%	30% prefer Blue
Combined, what fraction prefer NOT Blue?	Not Blue: Red+Green+Yellow = 7+5+3 = 15. Total = 24.	15/24 = 5/8

Multi-Step Example 3 — Limitations of Pictographs

A pictograph shows fruit sold. Scale: 1 apple symbol = 25 kg.

Month	Symbols	Calculation	Exact Kg
January	4.0	4 × 25	100 kg
February	3.5	3.5 × 25	87.5 kg
March	6.0	6 × 25	150 kg
April	2.75	2.75 × 25	68.75 kg

Critical Analysis Questions

Q1: Total fruit sold = 100 + 87.5 + 150 + 68.75 = 406.25 kg.

Q2: But fruit is measured in whole kg. The pictograph cannot represent fractional kg precisely → LIMITATION: pictographs are approximate.

Q3: Which month had sales CLOSEST to double February? Double of 87.5 = 175 kg. March = 150 kg (closest).

Q4: If the scale changed to 1 symbol = 50 kg, how many symbols for March (150 kg)? → 150 ÷ 50 = 3 symbols.

Q5: Is a pictograph the right choice for daily stock market data? → No. Too imprecise. Bar graph or line graph would be better.

INLINE ASSESSMENT — Module 2.1 (All Paths)

Path Routing for Inline Questions

PATH A: Starts at Q1 (Easy). Must answer correctly before seeing Q2. Remedial shown for every wrong answer. Cannot skip.

PATH B: Starts at Q1. All 3 questions attempted. Medium difficulty. One retry per wrong answer with remedial.

PATH C: Starts at Q2 directly (Q1 too basic). Q3 and Challenge Q4 attempted. Challenge unlocked if Q2 and Q3 both correct on first attempt.

Inline 2.1-Q1 [PATH A START — Easy] [Easy]

A frequency table: Dog=7, Cat=10, Fish=6, Rabbit=5. Which animal is MOST popular?

(A) Dog

(B) Cat

(C) Fish

(D) Rabbit

Hint: Most popular = highest frequency. Find the biggest number in the Frequency column.

Answer (B): Cat has frequency 10 — the largest value. Most popular = Cat.

Inline 2.1-Q2 [PATH A Q2 / PATH B Q1 / PATH C Q1] [Medium]

Pictograph of cars: one symbol = 50 cars. September shows 4 full symbols and 1 half symbol. How many cars?

(A) 200

(B) 225

(C) 250

(D) 175

Hint: Full symbols × value. Half symbol = half the value. Then add both.

Answer (B): 4 × 50 = 200. Half symbol = 25. Total = 200 + 25 = 225 cars.

Inline 2.1-Q3 [All Paths — PATH A Q3 / PATH B Q2 / PATH C Q2] [Medium]

Double bar graph: Hindi 2005-06=50, 2006-07=45. Maths 2005-06=30, 2006-07=70. Which subject showed a DECREASE in performance?

(A) Maths

(B) Hindi

(C) Both subjects

(D) Neither subject

Hint: Decrease = later value is LOWER than earlier value. Check each subject.

Answer (B): Hindi: 50 → 45 = decreased (−5). Maths: 30 → 70 = increased (+40). Answer: Hindi.

Inline 2.1-Q4 [PATH C CHALLENGE — Hard] [Hard]

A double bar graph shows boys vs girls marks in 5 subjects. Boys scores: Maths=40, Science=60, English=55, Hindi=70, Art=45. Girls

scores: Maths=55, Science=50, English=65, Hindi=65, Art=50. In how many subjects did BOYS outperform GIRLS?

(A) 1 subject

(B) 2 subjects

(C) 3 subjects

(D) 4 subjects

Hint: Outperform means score HIGHER. Compare boys vs girls in each subject one by one.

Answer (B): Maths: B40<G55. Science: B60>G50 ✓. English: B55<G65. Hindi: B70>G65 ✓. Art: B45<G50. Boys outperform in 2 subjects.

Remedial Content

Wrong on Q1: Scan the Frequency column. Find the LARGEST number. That row is the most popular. Here: 7, 10, 6, 5. Largest = 10 → Cat.

Wrong on Q2: Step 1 — count full symbols and multiply: 4 × 50 = 200. Step 2 — half symbol = scale ÷ 2 = 50 ÷ 2 = 25. Step 3 — add: 200 + 25 = 225.

Wrong on Q3: Decrease means the second (later) value is SMALLER than the first. Hindi went 50→45 so 45 < 50 → decreased. Do not confuse with Maths which went UP.

Wrong on Q4: Go subject by subject. Science: 60 > 50 (boys win). Hindi: 70 > 65 (boys win). All others: girls scored higher. Count = 2 subjects.

MODULE 2.2: Grouping Data & Histograms

PATH A | Score 0–40% | Full Foundational Content

ENGAGE — The Problem with Large Datasets

Scenario

Here are marks of 60 students:
21,10,30,22,33,5,37,12,25,42,15,39,26,32,18,27,8,41,23,30,19,35,29,3,44,31,17,28,24,38,11,20,36,45,16,29,13,40,22,34,7,48,27,30,9,43,18,26,2,47,14,31,24,38,20,27,6,35,19,42.Try to answer: What marks did most students score? How many scored above 40? A 60-row table is completely unreadable. The solution: GROUP data into class intervals.

EXPLAIN — Class Intervals (Every Term Defined)

Term	Definition	Example with 0–60 marks (width 10)
Class Interval	A range that groups values. Written as Lower–Upper	0–10, 10–20, 20–30, 30–40, 40–50, 50–60
Lower Class Limit	The smaller number at the start of the interval	0 is the lower limit of 0–10
Upper Class Limit	The larger number at the end of the interval	10 is the upper limit of 0–10
Class Width / Size	Upper Class Limit minus Lower Class Limit	10 − 0 = 10 for every interval
Class Mark (Midpoint)	(Lower Limit + Upper Limit) ÷ 2	(10+20) ÷ 2 = 15 is the class mark of 10–20
Boundary Convention	A value equal to the upper limit belongs to the NEXT (higher) class	The value 20 goes in 20–30, NOT in 10–20
Grouped Frequency	Number of data values that fall within a class interval	If 21 students scored in 20–30, that is the frequency

Why the Boundary Convention Matters

Suppose a student scored exactly 20. Class intervals are 10–20 and 20–30.
→ 20 belongs to 20–30 (the higher class), NOT 10–20.
This is a RULE, not a choice. It prevents any value from being counted twice.
The only exception: the final class interval is fully inclusive of its upper boundary.

Step-by-Step: Building a Grouped Frequency Table

Building the Table

7. Find the minimum (2) and maximum (48) values in the data.
8. Decide the class width. Here we choose 10. So intervals are: 0–10, 10–20, 20–30, 30–40, 40–50, 50–60.
9. Draw a table with columns: Class Interval, Tally Marks, Frequency.
10. Go through each data value ONE BY ONE. Decide which interval it belongs to (boundary convention applies).
11. Add a tally mark in the correct row. Strike through each value in the raw data as you count it.
12. Count all tallies in each row and write the frequency.
13. Verify: sum of all frequencies = 60 (total data points).

Class Interval	Tally Marks	Frequency	Interpretation
0 – 10	||	2	Very few scored this low
10 – 20	|||| |||||	10	Below average group
20 – 30	|||| |||| |||| |||| |	21	← LARGEST group — peak of distribution
30 – 40	|||| |||| |||| ||||	19	Second largest — above average
40 – 50	|||| ||	7	High scorers
50 – 60	|	1	Only 1 student scored full marks
TOTAL		60	Sum of frequencies = 60 ✓

What Is a Histogram?

Definition and Key Properties

A histogram is a bar graph for GROUPED NUMERICAL data (class intervals).

The most important rule: bars in a histogram have NO GAPS. They touch each other.

Why no gaps? Because class intervals are continuous — 0–10 ends at exactly 10, and 10–20 starts at exactly 10. There is no gap in the data, so there is no gap in the bars.

The horizontal axis shows the CLASS INTERVAL SCALE (numbers, not category names).

The vertical axis shows FREQUENCY.

Bar height = frequency of that class. Bar width = class width.

How to Draw a Histogram

14. Draw the horizontal (x) axis. Mark the class interval boundaries: 0, 10, 20, 30, 40, 50, 60.
15. Draw the vertical (y) axis. Mark frequency values: 0, 5, 10, 15, 20, 25.
16. For class 0–10 (freq=2): draw a bar from x=0 to x=10, height=2.
17. For class 10–20 (freq=10): draw a bar from x=10 to x=20, height=10. It touches the previous bar.
18. Continue for all classes. All bars must touch — NO gaps.
19. If the x-axis does not start from 0, draw a BREAK MARK (∧) to show the scale does not begin at zero.
20. Label both axes. Write a clear title.

Feature	Bar Graph	Histogram — KEY DIFFERENCE
Data Type	Categorical (names: Art, Red, City)	Numerical, grouped in class intervals
Gaps Between Bars	YES — always present	NO — bars always touching
Horizontal Axis	Category labels	Numerical scale (0, 10, 20, 30 …)
Can Bars Be Reordered?	Yes — any order	No — must follow numerical order
Bar Width	Equal, but represents a category	Equal and equals the class width
When to Use	Discrete categories	Continuous grouped measurements

PATH B | Score 40–70% | Standard Content — Class Intervals, Histograms, and Full Calculations

ENGAGE — From Unreadable Lists to Instant Insights

When you have 60 or 600 values in a dataset, a simple frequency table with one row per value is useless. Grouping data into class intervals transforms an unreadable dataset into a clear, analysable table in seconds.

EXPLAIN — Class Intervals and Grouped Frequency

Key Terms

Term	Meaning	Example (intervals of 10, marks 0–60)
Class Interval	A range grouping multiple values	20–30 groups all marks from 20 up to (but not including) 30
Lower Class Limit	The smaller number; marks ≥ this are included	20 is the lower limit of 20–30
Upper Class Limit	The larger number; marks < this are included (except final class)	30 is the upper limit of 20–30
Class Width	Upper limit − Lower limit	30 − 20 = 10
Class Mark	Midpoint of the interval: (Lower + Upper) ÷ 2	(20 + 30) ÷ 2 = 25
Boundary Convention	A value equal to the upper limit belongs to the HIGHER class	The value 30 goes into 30–40, NOT 20–30

The Boundary Convention — Worked Explanation

Why the Convention Exists and How to Apply It

Class intervals must cover every possible value exactly once — no value should be in two classes, and no value should be left out.

Convention: a value at the boundary (upper limit of one class = lower limit of the next) belongs to the HIGHER class.

Example: intervals are 10–20, 20–30, 30–40. A student scores exactly 20.

→ 20 = upper limit of 10–20. So 20 goes into 20–30 (the higher class).

→ A student scores exactly 30: goes into 30–40.

Exception: the FINAL class is fully closed at both ends. If final class is 50–60, then 60 is included in 50–60.

Building a Grouped Frequency Table — Full Walkthrough

Data: Marks of 60 Students

Raw marks:
21,10,30,22,33,5,37,12,25,42,15,39,26,32,18,27,8,41,23,30,19,35,29,3,44,31,17,28,24,38,11,20,36,45,16,29,13,40,22,34,7,48,27,30,9,43,18,26,2,47,14,31,24,38,20,27,6,35,19,42.

Class Interval	Tally	Frequency	Cumulative Frequency	Percentage
0–10	||	2	2	2/60 × 100 = 3.3%
10–20	|||| |||||	10	12	10/60 × 100 = 16.7%
20–30	|||| |||| |||| |||| |	21	33	21/60 × 100 = 35.0%
30–40	|||| |||| |||| ||||	19	52	19/60 × 100 = 31.7%
40–50	|||| ||	7	59	7/60 × 100 = 11.7%
50–60	|	1	60	1/60 × 100 = 1.7%

Class Interval	Tally	Frequency	Cumulative Frequency	Percentage
TOTAL		60	—	100%

Reading This Table

Modal class (most students): 20–30 (frequency 21, the highest).
Students scoring less than 30: cumulative frequency at 20–30 = 33 students.
Students scoring 30 or more: 60 − 33 = 27 students. (Or: 19+7+1 = 27.)
Students scoring in the middle 20-30 interval: 21 out of 60 = 35% of all students.

Histogram — Detailed Construction and Reading

What Makes a Histogram Different From a Bar Graph

Feature	Bar Graph	Histogram
Data type	Categorical (names, labels)	Numerical, grouped in class intervals
Gaps between bars	YES — must have gaps	NO — bars must touch
x-axis content	Category names (Art, Maths, Cities)	Numerical scale (0, 10, 20, 30 …)
Bar order	Can be rearranged freely	Must follow numerical order
Reason no gaps	N/A	Class intervals are continuous — 10–20 ends exactly where 20–30 begins

How to Construct the Histogram

Using the marks data: x-axis shows 0, 10, 20, 30, 40, 50, 60. y-axis shows frequency 0 to 25.

Class Interval	Bar from	Bar to	Height (Frequency)	Notes
0–10	x=0	x=10	2	Very short bar
10–20	x=10	x=20	10	Medium bar; starts exactly where previous ends
20–30	x=20	x=30	21	TALLEST bar — modal class
30–40	x=30	x=40	19	Second tallest
40–50	x=40	x=50	7	Short bar
50–60	x=50	x=60	1	Barely visible — very short

Worked Problems — Calculations from the Histogram

Question	Method	Answer
How many scored less than 20?	Add frequencies of classes with upper limit ≤ 20: 0-10 and 10-20	2 + 10 = 12 students
How many scored 40 or more?	Add classes with lower limit ≥ 40: 40-50 and 50-60	7 + 1 = 8 students
How many scored between 20 and 40?	Add classes 20-30 and 30-40	21 + 19 = 40 students
What fraction scored in 20-30?	21 out of 60	21/60 = 7/20

Question	Method	Answer
How many scored more than 30?	Add classes 30-40, 40-50, 50-60	19 + 7 + 1 = 27 students

PATH C | Score 70–100% | Advanced Content — Multi-step Problems and Comparative Analysis

ELABORATE — Advanced Histogram Analysis

You understand class intervals, boundary conventions, and how to read histograms. Path C extends to multi-step problems, working backwards from a histogram, comparing two distributions, and understanding the concept of a "modal class" and data spread.

Concept Review — Key Facts for Advanced Problems

Concept	Statement	Use in Problems
Modal Class	The class interval with the HIGHEST frequency	Identifies where most data is clustered
Cumulative Frequency	Running total from first class to current class	Answers "how many scored LESS THAN X?"
Boundary Convention	Value at upper limit → goes into HIGHER class	Placing values correctly when building the table
Class Mark	Midpoint = (Lower + Upper) ÷ 2	Used when an approximate single value is needed for the class
Fraction of Total	Frequency ÷ Total × 100 = Percentage	Answering "what % scored in class X–Y?"

Multi-Step Example 1 — Complete Analysis of a Histogram

Marks of 100 students: 0–10=3, 10–20=12, 20–30=25, 30–40=35, 40–50=18, 50–60=7. Total=100.

Question	Working	Answer
Modal class	Highest frequency = 35 → class 30–40	30–40
Students scoring < 20	3 + 12 = 15. As %: 15/100 = 15%	15 students (15%)
Students scoring ≥ 40	18 + 7 = 25. As %: 25/100 = 25%	25 students (25%)
Ratio: scored ≥ 30 to scored < 30	≥30: 35+18+7=60. <30: 3+12+25=40. Ratio = 60:40	3:2
Boundary: where does exactly 30 go?	Upper limit of 20–30 = 30 → belongs to 30–40	Class 30–40
If 5 more students join all scoring in 20–30, new freq?	25 + 5 = 30. New total = 105.	30 in 20–30 out of 105 total
Cutoff for top 25% to pass (i.e., bottom 75% fail)	75 students fail. Cumulative: 0–10=3, 10–20=15, 20–30=40, 30–40=75. 75th student is in 30–40.	Cutoff is within 30–40 interval

Multi-Step Example 2 — Comparing Two Histograms

Class A (50 students): 0–20=2, 20–40=15, 40–60=23, 60–80=8, 80–100=2.

Class B (50 students): 0–20=0, 20–40=8, 40–60=17, 60–80=18, 80–100=7.

Question	Class A	Class B	Conclusion
Modal class	40–60 (freq=23)	60–80 (freq=18)	Class A peaks lower
Students scoring ≥ 60	8+2=10 (20%)	18+7=25 (50%)	Class B has more high scorers
Students scoring < 40	2+15=17 (34%)	0+8=8 (16%)	Class A has more low scorers
Which class performed better?	Modal class 40–60	Modal class 60–80	Class B overall better

Key Insight — Distributions

Class A is "left-skewed" — more students clustered in lower ranges. Class B is "right-skewed" — more students clustered in higher ranges. The modal class tells you where the peak is.

Multi-Step Example 3 — Working Backwards

A histogram shows wages of workers. Total area (sum of frequency × width) = 200. Class width = 10 throughout.

Reading bar heights: 800–810=4, 810–820=6, 820–830=11, 830–840=9, 840–850=5, 850–860=3, 860–870=2. Total = 40 workers.

Question	Working	Answer
Workers earning less than 830?	4 + 6 + 11 = 21	21 workers
Workers earning 840 or more?	5 + 3 + 2 = 10	10 workers
Fraction earning in 820–830?	11/40	11/40 (do not simplify — 11 is prime)
If scale doubled (each bar reads ×2), total workers?	40 × 2 = 80	80 workers — error in reading scale
New class width if 7 bars merged into 3?	Approximate: combine 800–820, 820–840, 840–870	New widths: 20, 20, 30 (unequal)

INLINE ASSESSMENT — Module 2.2 (All Paths)

Path Routing

PATH A: Q1 (Easy) first. Full remedial with worked solution. Must answer correctly before Q2.
PATH B: Q1, Q2, Q3 — all Medium. One retry per question.
PATH C: Q2, Q3 only. Challenge Q4 unlocked if both correct on first attempt.

Inline 2.2-Q1 [PATH A START — Easy] [Easy]

Using the table (0-10=2, 10-20=10, 20-30=21, 30-40=19, 40-50=7, 50-60=1), how many students scored LESS THAN 20 marks?

(A) 2
(B) 10
(C) 12
(D) 21

Hint: Less than 20 = ALL classes whose upper limit is ≤ 20. Find which classes those are and add their frequencies.

Answer (C): 0–10 (freq=2) and 10–20 (freq=10). Total = 2+10 = 12 students.

Inline 2.2-Q2 [PATH B Q1 / PATH C Q1 — Medium] [Medium]

Workers wages grouped: 800–810=6, 810–820=3, 820–830=9, 830–840=8, 840–850=2. Which class has the MOST workers?

(A) 800–810

(B) 810–820

(C) 820–830

(D) 830–840

Hint: Most workers = highest frequency. Find the biggest number.

Answer (C): 820–830 has frequency 9 — the highest. Most workers earn in 820–830.

Inline 2.2-Q3 [All Paths — Medium] [Medium]

From the marks histogram (0-10=2, 10-20=10, 20-30=21, 30-40=19, 40-50=7, 50-60=1), how many students scored 40 marks or MORE?

(A) 7

(B) 8

(C) 9

(D) 6

Hint: 40 or more = classes 40–50 AND 50–60. Add both.

Answer (B): 40–50: 7 + 50–60: 1 = 8 students scored 40 or more.

Inline 2.2-Q4 [PATH C CHALLENGE — Hard] [Hard]

TV hours watched: 1hr=4, 2hr=8, 3hr=22, 4hr=32, 5hr=8, 6hr=6, 7hr=5. Total=85 students. What fraction watched MORE than 5 hours? Give in simplest form.

(A) 11/85

(B) 6/85

(C) 5/85

(D) 11/80

Hint: More than 5 hours = 6-hour group + 7-hour group. Then divide by total. Simplify.

Answer (A): More than 5 hrs: 6+5=11. Total=85. Fraction=11/85. GCD(11,85)=1. Already simplest.

Remedial Content

Wrong on Q1: "Less than 20" means every class whose values are below 20. That is 0–10 AND 10–20. You must ADD BOTH: 2+10=12. Many students mistakenly read only the 10–20 class.

Wrong on Q2: Scan the frequency column for the LARGEST number. Values are: 6, 3, 9, 8, 2. Largest=9. That is class 820–830.

Wrong on Q3: "40 or more" includes BOTH 40–50 (freq=7) AND 50–60 (freq=1). Do NOT forget the last class. 7+1=8.

Wrong on Q4: Step 1 — list which classes are "more than 5 hours": 6-hour (freq=6) and 7-hour (freq=5). Step 2 — add: 6+5=11. Step 3 — total students = 4+8+22+32+8+6+5=85. Step 4 — fraction = 11/85. Test if simplifiable: 11 is prime, 85=5×17. No common factor → 11/85 is simplest.

MODULE 2.3: Pie Charts / Circle Graphs

PATH A | Score 0–40% | Full Foundational Content — Pie Charts from Scratch

ENGAGE — Showing Everything in One Picture

Opening Problem

A family earns Rs 20,000 per month. They spend on: Food, Rent, Education, Clothes, Entertainment, and Savings. How can you show ALL expenses AND savings in a single picture where everything adds up to exactly 100%? A BAR GRAPH would show the amounts — but not the PROPORTION of the whole. A PIE CHART shows both: the actual value AND what fraction of the total each item is.

EXPLAIN — What Is a Pie Chart?

Building Understanding

A pie chart is a CIRCLE divided into SECTORS (wedge-shaped pieces, like slices of a pizza).
Each sector represents ONE CATEGORY of data.
The SIZE of a sector is PROPORTIONAL to the value it represents.
The whole circle = 360° = 100% of the data or total.
A bigger sector means a bigger value. A smaller sector means a smaller value.
The CENTRAL ANGLE of a sector is the angle at the centre of the circle for that slice.

THE FORMULA — How to Find the Central Angle

Central Angle Formula

Central Angle = (Value ÷ Total) × 360°

If given as percentage: Central Angle = (Percentage ÷ 100) × 360°Verify: ALL central angles must SUM to EXACTLY 360°

Step-by-Step: Drawing a Pie Chart

How to Make a Pie Chart

21. Find the TOTAL of all values. Add all categories.
22. For each category: Central Angle = (value ÷ total) × 360°. Work to nearest degree.
23. Check: sum of all angles must equal 360°.
24. Draw a circle. Using a protractor, measure and draw each sector angle from the centre.
25. Label each sector with the category name and percentage.
26. Give the pie chart a title.

Worked Example 1 — Music Preferences (Complete Calculation)

Survey of 400 people: Light=160, Folk=120, Classical=80, Semi-Classical=40. Total=400.
Light: (160÷400)×360° = (2/5)×360° = 144°
Folk: (120÷400)×360° = (3/10)×360° = 108°
Classical: (80÷400)×360° = (1/5)×360° = 72°
Semi-Classical: (40÷400)×360° = (1/10)×360° = 36°
Verify: 144+108+72+36 = 360° ✓

Category	Count	Fraction	Calculation	Central Angle
Light	160	160/400 = 2/5	(2/5)×360°	144°

Category	Count	Fraction	Calculation	Central Angle
Folk	120	120/400 = 3/10	(3/10)×360°	108°
Classical	80	80/400 = 1/5	(1/5)×360°	72°
Semi-Classical	40	40/400 = 1/10	(1/10)×360°	36°
TOTAL	400	1	Sum	360° ✓

Worked Example 2 — Bakery Sales

Items and sales (Rs): Ordinary Bread=320, Fruit Bread=80, Cakes=160, Biscuits=120, Others=40. Total=720.

Ordinary Bread: (320÷720)×360° = (4/9)×360° = 160°

Fruit Bread: (80÷720)×360° = (1/9)×360° = 40°

Cakes: (160÷720)×360° = (2/9)×360° = 80°

Biscuits: (120÷720)×360° = (1/6)×360° = 60°

Others: (40÷720)×360° = (1/18)×360° = 20°

Verify: 160+40+80+60+20 = 360° ✓

REVERSE CALCULATION — Reading a Pie Chart Backwards

What Is Reverse Calculation?

You are GIVEN the pie chart (percentages or angles) and asked to find ACTUAL VALUES.

Formula: Value = (Central Angle ÷ 360°) × Total

Or: Value = (Percentage ÷ 100) × Total

Alternative "chain method": if you know one sector's value and percentage, find the value of 1%, then scale to any other percentage.

Worked Example — Family Budget

Monthly income: Rs 20,000. Pie chart percentages: Food=40%, Rent=20%, Education=15%, Clothes=10%, Savings=15%.

Find Savings: (15÷100)×20,000 = Rs 3,000.

Find Clothes: (10÷100)×20,000 = Rs 2,000.

Find Food: (40÷100)×20,000 = Rs 8,000.

Chain method: Savings=15%=Rs3,000 → 1%=Rs200. Clothes(10%)=10×200=Rs2,000. Education(15%)=15×200=Rs3,000.

Verify: 8,000+4,000+3,000+2,000+3,000 = Rs 20,000 ✓

PATH B | Score 40–70% | Standard Content — Pie Charts: Full Formula, Multiple Examples, Reverse Calculation

ENGAGE — What Can Only a Pie Chart Show?

A bar graph shows quantities. A pie chart shows PROPORTION — what fraction of the total each part represents. When your question is "what percentage of the total is this?", a pie chart is the right choice. When your question is "how does this value compare in absolute terms?", use a bar graph.

EXPLAIN — The Central Angle Formula

Deriving the Formula

Why 360°?

A full circle = 360°. A pie chart uses a full circle to represent 100% of the data.

If one category is 1/4 of the total, it should occupy 1/4 of the circle = (1/4) × 360° = 90°.
If one category is 50% of the total, it occupies 50% of the circle = (50/100) × 360° = 180°.
General formula: Central Angle = (Category Value ÷ Total) × 360°
Equivalently: (Percentage ÷ 100) × 360°
Reverse: Value = (Angle ÷ 360°) × Total OR Value = (Percentage ÷ 100) × Total

Key Verification Rule

Must Check

ALL central angles must sum to exactly 360°. If they do not, recheck your calculations. A rounding error of 1–2° is acceptable due to rounding, but must be noted.

Worked Example 1 — Bakery Sales (Full Calculation)

A bakery sells 5 items. Total sales = Rs 720.

Item	Sales (Rs)	Fraction of Total	Calculation	Central Angle
Ordinary Bread	320	320/720 = 4/9	(4/9) × 360°	160°
Fruit Bread	80	80/720 = 1/9	(1/9) × 360°	40°
Cakes & Pastries	160	160/720 = 2/9	(2/9) × 360°	80°
Biscuits	120	120/720 = 1/6	(1/6) × 360°	60°
Others	40	40/720 = 1/18	(1/18) × 360°	20°
TOTAL	720	1	Sum	160+40+80+60+20 = 360° ✓

Worked Example 2 — Family Budget (Forward and Reverse)

Monthly income: Rs 20,000. Pie chart shows: Food=40%, Rent=20%, Education=15%, Clothes=10%, Savings=15%.

Forward Calculation — Finding Angles from Percentages

Category	Percentage	Calculation	Central Angle
Food	40%	(40/100) × 360°	144°
Rent	20%	(20/100) × 360°	72°
Education	15%	(15/100) × 360°	54°
Clothes	10%	(10/100) × 360°	36°
Savings	15%	(15/100) × 360°	54°
TOTAL	100%	Sum	144+72+54+36+54 = 360° ✓

Reverse Calculation — Finding Values from Percentages

Category	Percentage	Calculation	Amount Spent (Rs)
Food	40%	(40/100) × 20,000	Rs 8,000
Rent	20%	(20/100) × 20,000	Rs 4,000
Education	15%	(15/100) × 20,000	Rs 3,000
Clothes	10%	(10/100) × 20,000	Rs 2,000

Category	Percentage	Calculation	Amount Spent (Rs)
Savings	15%	(15/100) × 20,000	Rs 3,000
TOTAL	100%	Sum check	Rs 20,000 ✓

Worked Example 3 — Finding the Total from One Known Value

A pie chart shows music preferences. Classical = 10%. You are told 20 people liked Classical. Find the total number surveyed, and the number who preferred each other genre.

Step	Working	Result
Step 1: Find total	10% = 20 people → 1% = 2 → Total = 100% = 200 people	Total = 200
Folk = 30%	(30/100) × 200 = 60 people	60 people prefer Folk
Light = 40%	(40/100) × 200 = 80 people	80 people prefer Light
Semi-Classical = 20%	(20/100) × 200 = 40 people	40 people prefer Semi-Classical
Verification	20 + 60 + 80 + 40 = 200 ✓	Total checks out

Worked Example 4 — Election Results

360 voters chose: Candidate A=90, B=120, C=150. Total=360.

Candidate	Votes	Fraction	Central Angle	Shortcut When Total=360
A	90	90/360 = 1/4	(1/4)×360° = 90°	90 votes = 90° (1 vote = 1°)
B	120	120/360 = 1/3	(1/3)×360° = 120°	120 votes = 120°
C	150	150/360 = 5/12	(5/12)×360° = 150°	150 votes = 150°
TOTAL	360	1	360° ✓	—

PATH C | Score 70–100% | Advanced Content — Multi-variable and Comparative Pie Chart Problems

ELABORATE — Advanced Pie Chart Problems

You know the central angle formula and reverse calculation. Path C covers incomplete pie charts, multi-variable problems, comparing two pie charts, chain reasoning, and finding values when only partial information is given.

Multi-Step Example 1 — Incomplete Pie Chart

A family monthly budget pie chart shows: Food=120°, Rent=90°, Education=60°, Savings=?°. Monthly income = Rs 24,000.

Step	Calculation	Result
Find missing Savings angle	360° − 120° − 90° − 60° = 90°	Savings = 90°
Food amount	(120/360) × 24,000 = (1/3) × 24,000	Rs 8,000
Rent amount	(90/360) × 24,000 = (1/4) × 24,000	Rs 6,000

Step	Calculation	Result
Education amount	(60/360) × 24,000 = (1/6) × 24,000	Rs 4,000
Savings amount	(90/360) × 24,000 = (1/4) × 24,000	Rs 6,000
Verify	8,000 + 6,000 + 4,000 + 6,000 = Rs 24,000 ✓	Correct
Savings %	(90/360) × 100 = 25%	25% of income saved

Extension Questions

Q: If rent increases by Rs 2,000 next month (income = Rs 24,000 still), what is the new Rent angle?
New Rent = 6,000 + 2,000 = Rs 8,000. New angle = (8,000/24,000) × 360° = (1/3) × 360° = 120°.

Q: What must decrease to keep total = Rs 24,000? Some other category must reduce by Rs 2,000.

Q: If Savings stays at Rs 6,000, which single category absorbs the Rs 2,000 reduction? → One of Food, Education, or a new category.

Multi-Step Example 2 — Comparing Two Pie Charts

Budget Year 1 (income Rs 20,000): Food=40%, Rent=25%, Education=20%, Savings=15%.

Budget Year 2 (income Rs 24,000): Food=35%, Rent=22%, Education=25%, Savings=18%.

Category	Year 1 Amount	Year 2 Amount	Change in Rs	Change in %
Food	40%×20k = Rs 8,000	35%×24k = Rs 8,400	+Rs 400	Increased in amount, decreased in %
Rent	25%×20k = Rs 5,000	22%×24k = Rs 5,280	+Rs 280	Increased in amount, decreased in %
Education	20%×20k = Rs 4,000	25%×24k = Rs 6,000	+Rs 2,000	Biggest absolute increase
Savings	15%×20k = Rs 3,000	18%×24k = Rs 4,320	+Rs 1,320	Good improvement

Key Distinction

A category can DECREASE as a percentage of income but INCREASE in absolute amount if total income grew. Food went from 40% to 35% (percentage fell) but from Rs 8,000 to Rs 8,400 (amount rose). Never confuse percentage change with absolute change.

Multi-Step Example 3 — Chain Reasoning from Partial Data

A pie chart shows survey results. You know: Category A = 36°. Category B = 2 × A. Category C = 3 × A. Category D = remainder.

Step	Calculation	Result
A angle	Given	36°
B angle	2 × 36°	72°
C angle	3 × 36°	108°
D angle	360° − 36° − 72° − 108°	144°
As percentages: A	(36/360) × 100	10%
As percentages: B	(72/360) × 100	20%
As percentages: C	(108/360) × 100	30%

Step	Calculation	Result
As percentages: D	(144/360) × 100	40%
If B = 50 people, find total	B is 20% → 20% = 50 → 1% = 2.5 → Total = 250	250 people
D represents how many?	40% × 250	100 people

Multi-Step Example 4 — Sector Angle Worked Both Ways

A school has 1200 students. A pie chart shows language preferences: Hindi=45%, English=30%, Marathi=15%, Tamil=10%.

Language	Percentage	Central Angle	Number of Students
Hindi	45%	(45/100)×360° = 162°	(45/100)×1200 = 540
English	30%	(30/100)×360° = 108°	(30/100)×1200 = 360
Marathi	15%	(15/100)×360° = 54°	(15/100)×1200 = 180
Tamil	10%	(10/100)×360° = 36°	(10/100)×1200 = 120
TOTAL	100%	360° ✓	1200 ✓

Extension Questions

Q: How many more students prefer Hindi than Tamil? → 540 − 120 = 420 students.

Q: What angle represents students who prefer Hindi OR English? → 162° + 108° = 270°.

Q: If 60 more Tamil-speaking students join (total becomes 1260), what is the new Tamil angle? → New Tamil = 180. New angle = (180/1260) × 360° = (1/7) × 360° ≈ 51.4°.

Q: What is the ratio of Marathi to English speakers? → 180:360 = 1:2.

INLINE ASSESSMENT — Module 2.3 (All Paths)

Path Routing

PATH A: Q1 first (Medium). Remedial with step-by-step worked solution.

PATH B: Q1, Q2, Q3. One retry each. Medium to Hard.

PATH C: Q2 and Q3. Challenge Q4 unlocked if both correct first attempt.

Inline 2.3-Q1 [PATH A START — Medium] [Medium]

Music survey: Classical=10%. If 20 people liked Classical, how many were surveyed in TOTAL?

(A) 100

(B) 150

(C) 200

(D) 250

Hint: 10% of total = 20 people. Use unitary method: if 10% = 20, what is 100%?

Answer (C): If 10%=20, then 1%=2, so 100%=200 people surveyed total.

Inline 2.3-Q2 [PATH B Q1 / PATH C Q1 — Medium] [Medium]

360 people chose seasons: Summer=90, Rainy=120, Winter=150. Central angle for RAINY season?

(A) 90°
(B) 100°
(C) 120°
(D) 150°

Hint: Angle = (value ÷ total) × 360°. Total = 360.

Answer (C): (120÷360)×360° = (1/3)×360° = 120°.

Inline 2.3-Q3 [All Paths — Hard] [Hard]

Exam pie chart: Mathematics sector = 90°. Total marks in exam = 540. How many marks in Maths?

(A) 90
(B) 120
(C) 135
(D) 150

Hint: Marks = (angle ÷ 360°) × total. Convert angle to fraction first.

Answer (C): (90÷360)×540 = (1/4)×540 = 135 marks.

Inline 2.3-Q4 [PATH C CHALLENGE — Hard] [Hard]

Family pie chart: Rent=25%. If Rent = Rs 5,000, what is the total monthly income?

(A) Rs 15,000
(B) Rs 20,000
(C) Rs 25,000
(D) Rs 30,000

Hint: Rent=25% of total. If 25%=Rs5,000, find 100% using unitary method.

Answer (B): If 25%=Rs5,000, then 1%=Rs200. Total=100%=100×200=Rs20,000.

Remedial Content

Wrong on Q1: Classical = 10% of total. We are TOLD 10% = 20 people. Step 1: 10%=20 → 1%=20÷10=2. Step 2: Total=100%=100×2=200. The question asks for the TOTAL, not the fans of Classical.

Wrong on Q2: Formula — (value÷total)×360°. Here total=360 (people), so (120÷360)×360°=120°. When total happens to equal 360, each unit = 1 degree. Shortcut: 120 people = 120°.

Wrong on Q3: Step 1 — convert 90° to a fraction of the full circle: 90÷360=1/4. Step 2 — multiply that fraction by the total: (1/4)×540=135. Many students mistakenly try (90÷540)×360 — that gives the angle if you were MAKING the chart, not reading it.

Wrong on Q4: 25%=Rs5,000 → 1%=5,000÷25=Rs200 → 100%=200×100=Rs20,000. Always find 1% first in reverse calculation.

MODULE 2.4: Chance & Probability

PATH A | Score 0–40% | Full Foundational Content — Probability from Scratch

ENGAGE — What Is Chance?

Real-Life Situations

"It will probably rain tomorrow." — How likely? More than 50%? Less?

"I might win the lucky draw." — 1 out of 50 people entered. Very unlikely.

"I am certain to feel hungry by lunch time." — 100% likely.

"A new-born baby will be a boy." — Approximately 50% chance.

Probability is the mathematics of CHANCE — it gives us a way to measure and compare how likely events are.

EXPLAIN — Core Vocabulary (Every Term Defined)

Term	Definition	Example
Random Experiment	Experiment whose outcome CANNOT be predicted in advance, but all possible outcomes ARE known	Tossing a coin; rolling a die; drawing a ball
Outcome	One specific result of a single trial of the experiment	Getting 3 when a die is rolled
Sample Space (S)	The COMPLETE SET of all equally likely outcomes	Coin: {H, T}. Die: {1,2,3,4,5,6}. Two coins: {HH,HT,TH,TT}
Event (E)	One or more outcomes from the sample space that we are interested in	Event "even number on die" = {2,4,6}
Equally Likely	All outcomes have the SAME chance of happening	A fair die: 1,2,3,4,5,6 are all equally likely
Favourable Outcomes	Outcomes in the sample space that satisfy the event	For "prime on die": favourable = {2,3,5}

THE PROBABILITY FORMULA

The Formula

P(Event) = Number of Favourable Outcomes / Total Number of Equally Likely Outcomes

Range: P is ALWAYS between 0 and 1 ($0 \le P \le 1$).P=0 → Impossible. P=1 → Certain. P=0.5 → Equally likely to happen or not.Complement Rule: P(E) + P(not E) = 1, therefore P(not E) = 1 − P(E)

Step-by-Step: How to Solve Any Probability Problem

The Method

27. Identify the experiment (e.g., tossing a coin, rolling a die).
28. Write out the COMPLETE sample space. List ALL equally likely outcomes.
29. Identify the EVENT you are asked about.
30. List all outcomes in the sample space that SATISFY the event (favourable outcomes).
31. Count favourable outcomes. Count total outcomes.
32. Apply the formula: P = favourable ÷ total.

33. Simplify the fraction if possible.

Experiment	Sample Space	Event	Favourable Outcomes	Count	Total	P(Event)
Toss 1 coin	{H, T}	Getting Head	{H}	1	2	1/2
Roll a die	{1,2,3,4,5,6}	Getting a 4	{4}	1	6	1/6
Roll a die	{1,2,3,4,5,6}	Getting even	{2,4,6}	3	6	1/2
Roll a die	{1,2,3,4,5,6}	Getting prime	{2,3,5}	3	6	1/2
Roll a die	{1,2,3,4,5,6}	Getting > 4	{5,6}	2	6	1/3
Bag: 4R,2Y	{R,R,R,R,Y,Y}	Red ball	{R,R,R,R}	4	6	2/3
Bag: 4R,2Y	{R,R,R,R,Y,Y}	NOT red	{Y,Y}	2	6	1/3
Slips 1–10	{1,2,3,4,5,6,7,8,9,10}	Greater than 6	{7,8,9,10}	4	10	2/5
Slips 1–10	{1,2,...,10}	1-digit number	{1,2,3,4,5,6,7,8,9}	9	10	9/10
2 coins	{HH,HT,TH,TT}	Exactly 1 Head	{HT,TH}	2	4	1/2
2 coins	{HH,HT,TH,TT}	At least 1 Head	{HH,HT,TH}	3	4	3/4

Critical Warning — Do NOT Guess Primes

1 is NOT a prime number (it has only 1 factor). Primes = {2,3,5,7,11…} — exactly 2 factors: 1 and itself.On a die: primes = {2,3,5}. NOT {1,2,3,5}. Count = 3, NOT 4.

PATH B | Score 40–70% | Standard Content — Probability: Full Formula, Multiple Worked Examples

ENGAGE — From Guessing to Measuring

Before this lesson: "I think it might rain" or "I might roll a six." After this lesson: "There is a 40% chance of rain" or "The probability of rolling a six is 1/6 = 16.7%." Probability transforms vague guesses into precise, comparable measurements.

EXPLAIN — Core Vocabulary and Formula

Essential Terms

Term	Definition	Example
Random Experiment	Outcome cannot be predicted, but all possible outcomes are known	Rolling a die: outcome unknown, but must be 1,2,3,4,5, or 6
Sample Space	Complete list of ALL equally likely outcomes	Die: {1,2,3,4,5,6}. Coin: {H,T}. Two coins: {HH,HT,TH,TT}
Event	One or more outcomes we are interested in	"Rolling an even number" = {2,4,6}
Favourable Outcomes	Outcomes in the sample space that satisfy the event	For "even number": {2,4,6} = 3 favourable
P(E)	Probability of event E	P(even) = 3/6 = 1/2

Term	Definition	Example
Complement P(not E)	Probability that E does NOT happen	P(not even) = 1 − P(even) = 1 − 1/2 = 1/2

The Formula

P(Event) = Number of Favourable Outcomes ÷ Total Number of Equally Likely OutcomesRange: 0 ≤ P ≤ 1. P=0 means impossible. P=1 means certain.Complement Rule: P(E) + P(not E) = 1

Worked Example 1 — Balls in a Bag

A bag contains 3 Red, 2 Blue, 1 Green. Total = 6 balls.

Event	Favourable Outcomes	Count	Total	P(Event)	Simplified
Drawing Red	{ R, R, R }	3	6	3/6	1/2
Drawing Blue	{ B, B }	2	6	2/6	1/3
Drawing Green	{ G }	1	6	1/6	1/6
NOT Red (Blue or Green)	{ B, B, G }	3	6	3/6	1/2
Red OR Blue	{ R,R,R,B,B }	5	6	5/6	5/6

Using the Complement Rule

P(NOT Red) can be found TWO ways:Method 1: Count not-red balls → Blue(2) + Green(1) = 3. P = 3/6 = 1/2.Method 2: P(NOT Red) = 1 − P(Red) = 1 − 3/6 = 1 − 1/2 = 1/2. ✓ Both methods agree.

Worked Example 2 — Numbered Slips

Slips numbered 1 to 10. One drawn at random. Total = 10.

Event	List Favourable Outcomes	Count	P(Event)
Getting 6	{ 6 }	1	1/10
Greater than 6	{ 7, 8, 9, 10 }	4	4/10 = 2/5
Less than 6	{ 1, 2, 3, 4, 5 }	5	5/10 = 1/2
Between 3 and 8 (exclusive)	{ 4, 5, 6, 7 }	4	4/10 = 2/5
Even number	{ 2, 4, 6, 8, 10 }	5	5/10 = 1/2
Prime number	{ 2, 3, 5, 7 }	4	4/10 = 2/5
1-digit number	{ 1, 2, 3, 4, 5, 6, 7, 8, 9 }	9	9/10
NOT 1-digit (i.e., 10)	{ 10 }	1	1/10

Worked Example 3 — Two Coins Tossed

When two coins are tossed, there are 4 equally likely outcomes. The sample space is {HH, HT, TH, TT}.

IMPORTANT: HT and TH are DIFFERENT outcomes. H on the first coin and T on the second is not the same as T on the first and H on the second.

Event	Favourable Outcomes	Count	P(Event)
Exactly 2 Heads (HH)	{ HH }	1	1/4
Exactly 1 Head	{ HT, TH }	2	2/4 = 1/2
Exactly 0 Heads (TT)	{ TT }	1	1/4
At least 1 Head	{ HH, HT, TH }	3	3/4
At most 1 Head (0 or 1 Head)	{ TT, HT, TH }	3	3/4
NOT getting 2 Heads (= at most 1)	Complement: 1 − P(HH) = 1 − 1/4	—	3/4

Shortcut Using Complement

"At least 1 Head" = 1 − P(no heads) = 1 − P(TT) = 1 − 1/4 = 3/4.This shortcut works whenever "at least one" problems have a simple complement (i.e., the "none" case is easy to calculate).

Worked Example 4 — Die Rolls

A single die is rolled. Sample space = {1, 2, 3, 4, 5, 6}. Total = 6 equally likely outcomes.

Event	Favourable Outcomes	Count	P(Event)
Even number	{ 2, 4, 6 }	3	3/6 = 1/2
Odd number	{ 1, 3, 5 }	3	3/6 = 1/2
Prime number	{ 2, 3, 5 }	3	3/6 = 1/2
NOT prime	{ 1, 4, 6 }	3	3/6 = 1/2
Greater than 4	{ 5, 6 }	2	2/6 = 1/3
Perfect square	{ 1, 4 }	2	2/6 = 1/3
Divisible by 3	{ 3, 6 }	2	2/6 = 1/3
Getting 7	{ } (impossible)	0	0/6 = 0

PATH C | Score 70–100% | Advanced Content — Compound Events, Expected Frequency, and Analysis

ELABORATE — Compound Events and Advanced Analysis

You know basic probability and the complement rule. Path C covers: compound events (AND / OR), multi-ball scenarios, linking probability to expected frequency, and problems requiring careful listing of sample spaces.

Concept Review — Rules for Advanced Problems

Concept	Rule	Example
AND condition	Both conditions must be satisfied. List intersection of both sets.	P(>3 AND <8) on slips 1–10 = {4,5,6,7} = 4/10
OR condition	Either condition satisfied (or both). List union.	P(prime OR even) on die = {2,3,4,5,6} = 5/6
Complement	P(not E) = 1 − P(E). Use when "not" is simpler.	P(not prime) = 1 − 3/6 = 1/2

Concept	Rule	Example
Expected Frequency	E(event) = P(event) × total trials	P(H) = 1/2. 100 tosses → expect 50 Heads
Listing carefully	Never guess counts. Always list ALL favourable outcomes first, then count.	"Prime on die": list {2,3,5} → count = 3, NOT 4

Multi-Step Example 1 — AND Conditions

Slips numbered 1–10. One slip drawn. Find probabilities for compound conditions.

Event (AND condition)	Method: List numbers satisfying BOTH	Favourable	P(Event)
> 3 AND < 8	Numbers > 3: {4,5,6,7,8,9,10}. Numbers < 8: {1,2,3,4,5,6,7}. Intersection: {4,5,6,7}	4	4/10 = 2/5
Even AND > 5	Even: {2,4,6,8,10}. >5: {6,7,8,9,10}. Intersection: {6,8,10}	3	3/10
Odd AND < 4	Odd: {1,3,5,7,9}. <4: {1,2,3}. Intersection: {1,3}	2	2/10 = 1/5
Prime AND even	Prime: {2,3,5,7}. Even: {2,4,6,8,10}. Intersection: {2}	1	1/10
Prime AND > 6	Prime: {2,3,5,7}. >6: {7,8,9,10}. Intersection: {7}	1	1/10

Multi-Step Example 2 — Complement Chain

A bag has 4 Red, 3 Blue, 2 Green, 1 Yellow. Total = 10 balls.

Event	Direct Method	Complement Method	P(Event)
Red	Favourable: 4	1 − P(not red) = 1 − 6/10	4/10 = 2/5
Not Red	Favourable: B+G+Y = 6	1 − P(Red) = 1 − 4/10	6/10 = 3/5
Not Red AND Not Blue	Favourable: G+Y = 3	1 − P(Red or Blue) = 1 − 7/10	3/10
Red OR Blue	Favourable: 4+3 = 7	1 − P(not red AND not blue) = 1 − 3/10	7/10
Not Green	Favourable: R+B+Y = 8	1 − P(Green) = 1 − 2/10	8/10 = 4/5

Extension: If 2 Green Balls Added

New bag: 4R, 3B, 4G, 1Y. New total = 12.P(Green) = 4/12 = 1/3.P(Red or Green) = (4+4)/12 = 8/12 = 2/3.P(NOT Blue) = 1 − P(Blue) = 1 − 3/12 = 1 − 1/4 = 3/4.

Multi-Step Example 3 — Two-Coin Detailed Analysis

Sample space: {HH, HT, TH, TT}. Four equally likely outcomes.

Event	Outcomes Satisfying Event	Count	P	Using Complement?
Exactly 2 Heads	{HH}	1	1/4	P(not HH) = 3/4
Exactly 1 Head	{HT, TH}	2	2/4 = 1/2	—

Event	Outcomes Satisfying Event	Count	P	Using Complement?
Exactly 0 Heads	{TT}	1	1/4	P(at least 1) = 3/4
At least 1 Head	{HH, HT, TH}	3	3/4	= 1 − P(TT) = 1 − 1/4
At most 1 Head	{TT, HT, TH}	3	3/4	= 1 − P(HH) = 1 − 1/4
First coin is Head	{HH, HT}	2	2/4 = 1/2	Independent of second coin
Both coins same	{HH, TT}	2	2/4 = 1/2	—

Key Question

Are "at least 1 Head" and "at most 1 Head" complementary? NO — they both have probability 3/4 and they OVERLAP (at "exactly 1 Head"). Complementary events must sum to 1 and have no overlap. Here: 3/4 + 3/4 = 6/4 ≠ 1.

Multi-Step Example 4 — Expected Frequency

Probability predicts EXPECTED outcomes over many trials, not guaranteed results for any single trial.

Experiment	P(Event)	Number of Trials	Expected Frequency	Formula
Toss a coin: Heads	1/2	100 tosses	50 Heads expected	(1/2) × 100 = 50
Roll a die: get 6	1/6	60 rolls	10 sixes expected	(1/6) × 60 = 10
Roll a die: prime	1/2	60 rolls	30 prime rolls	(1/2) × 60 = 30
Draw Red ball: bag 4R,6B	4/10	50 draws	20 Red expected	(4/10) × 50 = 20
Two coins: at least 1 Head	3/4	80 tosses	60 times expected	(3/4) × 80 = 60

Important Note

Expected frequency is a PREDICTION — actual results will vary.

Example: Toss a coin 10 times. You might get 4 Heads, not 5. But over 10,000 tosses, you will get close to 5,000 Heads.

The law of large numbers: as trials increase, actual frequency gets closer to expected frequency.

If a coin gives 8 Heads in 10 tosses, it does NOT mean the coin is unfair — small samples vary widely.

INLINE ASSESSMENT — Module 2.4 (All Paths)

Path Routing

PATH A: Q1 (Easy) and Q2 (Easy). Must get Q1 correct before Q2.

PATH B: Q2, Q3, and Q4. One retry with remedial.

PATH C: Q3 and Q4. Challenge Q5 unlocked if both correct.

Inline 2.4-Q1 [PATH A START — Easy] [Easy]

Bag contains 3 Red, 2 Blue, 1 Green. One ball drawn at random. What is P(Red)?

(A) 1/6
(B) 2/6
(C) 3/6
(D) 4/6

Hint: Count red balls (favourable). Count all balls (total). Apply formula.

Answer (C): 3 red out of 6 total. P(Red) = 3/6 = 1/2.

Inline 2.4-Q2 [PATH A Q2 / PATH B Q1 — Easy] [Easy]

A die is thrown once. What is P(prime number)?

(A) 1/6
(B) 2/6
(C) 3/6
(D) 4/6

Hint: List ALL numbers on a die. Which ones are prime (exactly 2 factors)?

Answer (C): Primes on die: {2,3,5} = 3. P(prime) = 3/6 = 1/2.

Inline 2.4-Q3 [PATH B Q2 / PATH C Q1 — Medium] [Medium]

Slips numbered 1–10. One drawn. What is P(number greater than 6)?

(A) 3/10
(B) 4/10
(C) 5/10
(D) 6/10

Hint: List all numbers from 1–10 that are STRICTLY greater than 6.

Answer (B): Numbers > 6: {7,8,9,10} = 4 favourable. P = 4/10 = 2/5.

Inline 2.4-Q4 [PATH B Q3 / PATH C Q2 — Hard] [Hard]

Two coins tossed. Sample space: {HH, HT, TH, TT}. What is P(at least one Head)?

(A) 1/4
(B) 2/4
(C) 3/4
(D) 4/4

Hint: "At least one" means 1 OR more. List which outcomes have at least 1 Head.

Answer (C): HH ✓, HT ✓, TH ✓, TT ✗. 3 favourable out of 4. P = 3/4.

Inline 2.4-Q5 [PATH C CHALLENGE — Hard] [Hard]

Slips 1–10. P(number > 3 AND number < 8) = ?

(A) 3/10
(B) 4/10
(C) 5/10
(D) 2/10

Hint: List numbers satisfying BOTH conditions: greater than 3 AND less than 8.

Answer (B): {4,5,6,7} = 4 numbers. P = 4/10 = 2/5.

Remedial Content

Wrong on Q1: Count only the RED balls for favourable: 3. Count ALL balls for total: 3+2+1=6. P=3/6. Do not include other colours in the top of the fraction.

Wrong on Q2: List die faces: 1,2,3,4,5,6. Is 1 prime? NO (1 factor only). Is 4 prime? NO (4=2×2). Is 6 prime? NO (6=2×3). Primes: {2,3,5} = 3. P=3/6=1/2.

Wrong on Q3: "Greater than 6" is STRICTLY greater — does NOT include 6 itself. Numbers: 7,8,9,10. Count=4. P=4/10.

Wrong on Q4: Write sample space: {HH,HT,TH,TT}. Mark each: HH has 2 heads ✓, HT has 1 ✓, TH has 1 ✓, TT has 0 ✗. Count ticks = 3. P=3/4. Or use complement: P(no heads)=P(TT)=1/4. At-least-1 = 1−1/4=3/4.

Wrong on Q5: You need BOTH conditions: >3 AND <8. List numbers > 3: {4,5,6,7,8,9,10}. Then remove those that are NOT < 8: remove 8,9,10. Remaining: {4,5,6,7}. Count=4. P=4/10.

Section 3: Post-Test — 10 Questions

Post-Test Design

Covers all 4 sub-modules with same 13 skill tags as the Pre-Test.
Difficulty: 3 Easy, 5 Medium, 2 Hard.
Learning Gain = Post% − Pre% per sub-topic.
Gain < 10% → "Re-attempt module" triggered. Gain ≥ 30% → "Excellent Progress" badge.

Post-Q1 [Easy]

Frequency table: Dog=7, Cat=10, Fish=6, Rabbit=5, Cow=2 (Total=30). What FRACTION prefer Fish?

(A) 5/30
(B) 6/30
(C) 7/30
(D) 10/30

Hint: Fraction = Fish frequency ÷ total.

Answer (B): 6 out of 30 → 6/30. Skill: frequency_table, fraction_percent

Post-Q2 [Easy]

Bar graph enrolment: 2007-08=325, 2003-04=100. By how much did enrolment increase?

(A) 100
(B) 225
(C) 325
(D) 200

Hint: Increase = final − starting value.

Answer (B): 325 − 100 = 225 students. Skill: bar_graph_read

Post-Q3 [Easy]

Class interval 20–30 in a frequency table. What is the width of this class interval?

(A) 5
(B) 10
(C) 20
(D) 30

Hint: Width = upper limit − lower limit.

Answer (B): 30 − 20 = 10. Skill: class_interval_basics

Post-Q4 [Medium]

A student scores exactly 30. Intervals: 20–30 and 30–40. Which interval does 30 belong to?

(A) 20–30
(B) 30–40
(C) Both intervals
(D) Neither interval

Hint: Boundary convention: belongs to the HIGHER class.

Answer (B): 30 belongs to 30–40. Skill: class_interval_basics

Post-Q5 [Medium]

A histogram has no gaps between bars because:

(A) Data has no gaps

(B) Class intervals are continuous — one ends exactly where the next begins

(C) Drawn with a ruler

(D) Each bar = different category

Hint: Think: what makes class intervals continuous?

Answer (B): 10–20 ends where 20–30 begins — continuous. Skill: histogram_features

Post-Q6 [Medium]

36 students: Blue=18, Green=9, Red=6, Yellow=3. Central angle for YELLOW?

(A) 15°

(B) 30°

(C) 45°

(D) 60°

Hint: Angle = (3 ÷ 36) × 360°.

Answer (B): (3/36)×360° = (1/12)×360° = 30°. Skill: central_angle_calc

Post-Q7 [Medium]

Pie chart: Education=15%. Monthly income = Rs 20,000. Amount on Education?

(A) Rs 2,000

(B) Rs 3,000

(C) Rs 4,000

(D) Rs 5,000

Hint: Education = 15% of Rs 20,000.

Answer (B): (15/100)×20,000 = Rs 3,000. Skill: pie_read_reverse

Post-Q8 [Medium]

Hostel: Hindi=40, English=12, Marathi=9, Tamil=7, Bengali=4 (Total=72). Central angle for HINDI?

(A) 160°

(B) 180°

(C) 200°

(D) 220°

Hint: (40 ÷ 72) × 360°. Simplify first.

Answer (C): (40/72)×360° = (5/9)×360° = 200°. Skill: central_angle_calc

Post-Q9 [Hard]

Slips 1–10. P(number BOTH > 3 AND < 8)?

(A) 4/10

(B) 3/10

(C) 5/10

(D) 2/10

Hint: List: greater than 3 AND less than 8.

Answer (A): Numbers: {4,5,6,7} = 4. P = 4/10. Skill: prob_compound

Post-Q10 [Hard]

Bag: 4 Red, 2 Yellow. E1=P(red), E2=P(yellow). Which is correct?

(A) P(E1)+P(E2)=2

(B) P(E1)+P(E2)=1

(C) P(E1)=P(E2)

(D) P(E1)<P(E2)

Hint: E1 and E2 cover all outcomes — they are complementary.

Answer (B): P(red)=4/6, P(yellow)=2/6. Sum=1. Skill: prob_complement

Section 4: System Simulation — 3 Learner Profiles

Each simulation documents the full journey: Pre-Test → content path shown → inline questions → Post-Test result.

LEARNER A: Priya — Score 25% → PATH A Full Content

Field	Value
Pre-Test Score	2/8 = 25% → PATH A for ALL 4 modules
Correct	PRE 2 (pictograph), PRE 6 (spinner)
Content	Full foundational: all concepts, all examples, complete step-by-step

ST EP1	SYSTEM	Pre-Test 8 Qs shown. Priya completes all.	→ Score=25% → PATH A all modules
ST EP2	MODULE 2.1	Full Path A: Engage story → raw data → tally steps → frequency table → bar/double bar/pictograph rules.	→ Spends 15 mins. Time flag: STRUGGLING
ST EP3	PRIYA	Q1 (Easy): chooses Dog instead of Cat.	→ WRONG → Remedial shown: "Largest frequency = most popular"
ST EP4	PRIYA	Q1 retry: chooses Cat. Correct.	→ Attempt 2 correct. Difficulty stays Easy.
ST EP5	MODULE 2.2	Full Path A: class interval table, boundary convention, 6-step table build, 7-step histogram draw.	→ Reads all pages. Q1 (Easy): correct first try.
ST EP6	MODULE 2.3	Full Path A: formula derivation, 2 worked examples, reverse calculation chain method.	→ Q1 (Medium): correct using unitary method.
ST EP7	MODULE 2.4	Full Path A: all vocabulary defined, 11-row worked example table, complement rule.	→ Q1 easy (red ball): correct. Q2 (prime): wrong → remedial.
ST EP8	POST-TEST	Scores 6/10 = 60%.	→ Gain = +35%. Badge: Excellent Progress!
ST EP9	SYSTEM	Weak: histogram_calc, prob_complement. Strong: bar_graph, pie_charts.	→ Recommendation: Re-attempt Modules 2.2 and 2.4

LEARNER B: Arjun — Score 50% → Mixed Paths

Field	Value
Pre-Test Score	4/8 = 50%
Module 2.1	Correct: Q1,Q2,Q3,Q5 → PATH C (advanced)
Module 2.2	Incorrect: Q5,Q8 → PATH A (full foundational)

Field	Value
Module 2.3	Correct: Q4,Q7 → PATH C (advanced)
Module 2.4	Correct: Q6 but low overall → PATH B (standard)

ST EP1	MODULE 2.1	PATH C: complex 3-dataset comparison example. Reads in 3 min.	→ 2.1-Q2 and Q3 correct → Challenge Q4 unlocked
ST EP2	ARJUN	Challenge Q4 (Hard — boys vs girls, 5 subjects): identifies 2 subjects.	→ Correct. Module 2.1: mastered.
ST EP3	MODULE 2.2	PATH A full content: boundary convention, step-by-step table, histogram drawing rules.	→ Arjun studies carefully for 10 minutes.
ST EP4	ARJUN	2.2-Q2 correct. 2.2-Q3 correct. Module 2.2: 2/2.	→ Histogram skill improved significantly.
ST EP5	MODULE 2.3	PATH C: incomplete pie chart, two-chart comparison, chain reasoning.	→ 2.3-Q3 wrong (angle→marks confusion) → remedial.
ST EP6	ARJUN	Retry 2.3-Q3: uses (90÷360)×540=135. Correct.	→ Challenge Q4 (find total from Rent%): correct.
ST EP7	MODULE 2.4	PATH B: 4 worked examples. 2.4-Q2,Q3,Q4 attempted.	→ Q4 (at-least-1-head) uses complement → correct.
ST EP8	POST-TEST	Scores 7/10 = 70%.	→ Gain=+20%. Weak: pie_chart_reverse_calc.

LEARNER C: Sneha — Score 88% → Mostly PATH C

Field	Value
Pre-Test Score	7/8 = 88%
Missed	PRE 8 (class interval convention) → Module 2.2 gets PATH B
All others	PATH C advanced content

ST EP1	MODULE 2.1	PATH C: 3-dataset double bar example, pattern analysis, average calculation.	→ Reads 2 min. RUSHING flag → "Please review before continuing".
ST EP2	SNEHA	2.1-Q2,Q3 correct → Challenge Q4 (boys vs girls 5 subjects).	→ Q4 correct. Module 2.1: mastered.
ST EP3	MODULE 2.2	PATH B standard: boundary convention example, histogram construction, range calculations.	→ 5 minutes. Q2,Q3 both correct on first attempt.
ST EP4	MODULE 2.3	PATH C: incomplete pie chart, two-year budget comparison, chain reasoning.	→ Q2,Q3 correct. Challenge Q4 (find income from Rent%): correct.

ST EP5	**MODULE 2.4**	PATH C: compound AND conditions, complement chains, two-coin analysis.	**→ Q3,Q4,Q5 (compound condition) all correct.**
ST EP6	**POST-TEST**	Scores 10/10 = 100%.	**→ Gain=+12%. All 13 skill tags: MASTERED.**
ST EP7	**SYSTEM**	mastery_status = mastered for all sub-topics. Extension problems unlocked.	**→ Optional: advanced probability (expected frequency, multi-stage).**

Appendix A: Learner Model Data Structure

Field	Type	Description
learner_id	UUID	Unique identifier per learner
subtopic_id	Enum	data_org | histograms | pie_charts | probability
pretest_score	Float 0–1	Fraction of pre-test questions correct for this sub-topic
posttest_score	Float 0–1	Post-test fraction correct
learning_gain	Float	posttest_score − pretest_score
content_path	Enum	A (full) | B (standard) | C (advanced)
mastery_status	Enum	not_started | in_progress | mastered | needs_review
remedial_shown	Integer	Times remedial triggered in this module
challenge_unlocked	Boolean	True if first 3 inline Qs correct on first attempt
time_flag	Enum	normal | rushing | struggling | hint_triggered
attempt_number	Integer	1=first try; 2=retry after remedial

Appendix B: Skill Tag Mapping

Skill Tag	Sub-Module	Pre-Test Q	Inline Q	Post-Test Q
bar_graph_read	2.1	Q1	2.1-Q3	Post-Q2
pictograph_read	2.1	Q2	2.1-Q2	—
double_bar_read	2.1	Q3	2.1-Q3,Q4	—
frequency_table	2.1	Q5	2.1-Q1	Post-Q1
class_interval_basics	2.2	Q8	2.2-Q2	Post-Q3,Q4
histogram_features	2.2	—	2.2-Q2	Post-Q5
histogram_calc	2.2	—	2.2-Q1,Q3,Q4	—
fraction_percent	2.3	Q4	—	Post-Q1
central_angle_calc	2.3	Q7	2.3-Q2	Post-Q6,Q8
pie_read_reverse	2.3	Q7	2.3-Q1,Q3,Q4	Post-Q7
prob_basic	2.4	Q6	2.4-Q1,Q2	Post-Q9
prob_complement	2.4	—	2.4-Q4	Post-Q10
prob_compound	2.4	—	2.4-Q5	Post-Q9